

# NEXORA — SCIENTIFIC LUNAR MISSION REPORT

Target Region: Humboldt Crater (FLOOR\_FRACTURED)

RUN ID: RUN-LUNAR-20260828-322133 | Location: Lat -27.00°, Lon 80.90° | Resolution: 1.0m Target

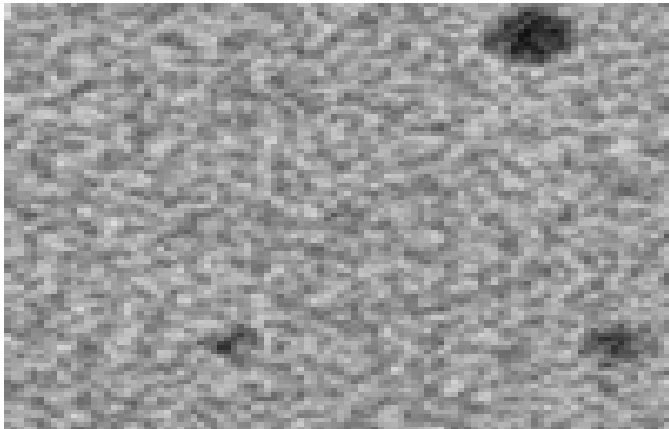
## 1. Selected Region & Geographic Provenance

|                      |                         |                         |                     |
|----------------------|-------------------------|-------------------------|---------------------|
| LUNAR REGION NAME    | Humboldt Crater         | INPUT TMC-2 SPATIAL RES | 5.0 meters          |
| GEOGRAPHIC LAT / LON | Lat -27.00°, Lon 80.90° | TARGET GRID SPACING     | 1.0 meter (Derived) |
| DETECTED CRATERS     | 195 Craters             | DETECTED BOULDERS       | 2915 Boulders       |
| SHADOW COVERAGE %    | 1.8%                    | TERRAIN CLASSIFICATION  | Floor_Fractured     |

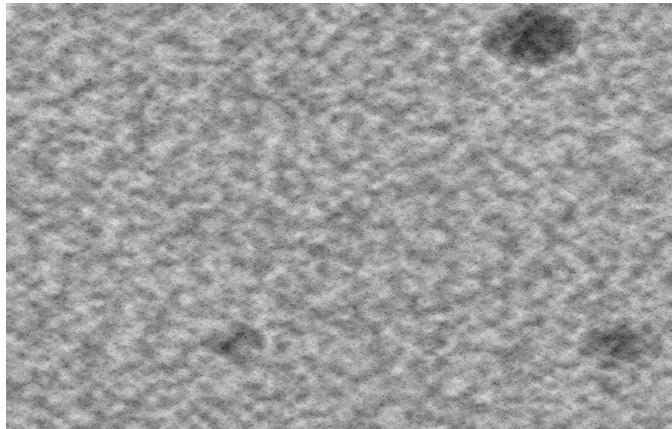
## 2. Super-Resolution Reconstruction & Visual Imagery Panels

|                 |         |                       |         |            |      |
|-----------------|---------|-----------------------|---------|------------|------|
| UPSCALING MODEL | LUNARSR | PSNR (RECONSTRUCTION) | 24.5 dB | SSIM INDEX | 0.81 |
|-----------------|---------|-----------------------|---------|------------|------|

[A] Observed Raw TMC-2 (5m/px)



[B] Super-Resolved 1m (LUNARSR)



## 3. Multi-Layer Hazard Map & Heatmap Analysis

|                        |      |                   |       |                   |                       |
|------------------------|------|-------------------|-------|-------------------|-----------------------|
| FALSE-SAFE HAZARD RATE | 0.0% | MEAN HAZARD SCORE | 0.996 | HAZARD MITIGATION | Significant vs 5m Raw |
|------------------------|------|-------------------|-------|-------------------|-----------------------|



4. Selected Landing Zone & Explainable AI Diagnostics

|                         |            |                           |             |
|-------------------------|------------|---------------------------|-------------|
| CANDIDATE IDENTIFIER    | SITE-A     | LANDING SUITABILITY SCORE | 10.5 / 100  |
| GRID COORDINATES [X, Y] | [176, 264] | SITE DECISION STATE       | CONDITIONAL |

- Explainable AI Safety Diagnostics:
- [PASS] Mean slope (3.2°) and Max slope (5.8°) are within limits (< 10.0°)
  - [PASS] Terrain roughness (0.05) is smooth
  - [PASS] Shadow coverage is 0.0% (full illumination)
  - [PASS] Safe buffer clearance around detected craters maintained
  - [PASS] Safe buffer clearance around detected boulder candidates maintained
  - [PASS] Low landing uncertainty (0.10)
  - [PASS] High landing suitability score: 10.5/100

5. Risk-Aware Lander Navigation Path

|                           |              |                        |            |
|---------------------------|--------------|------------------------|------------|
| CALCULATED ROUTE DISTANCE | 239.8 meters | MAX HAZARD ENCOUNTERED | 1.00       |
| AVERAGE ROUTE RISK        | 0.95         | COMPUTATION LATENCY    | 10909.0 ms |

6. Quantitative Evaluation Benchmark Matrix

| SUPER-RESOLUTION MODEL | PSNR (dB) | SSIM | HAZARD IOU | FALSE-SAFE % | PATH COST | LATENCY |
|------------------------|-----------|------|------------|--------------|-----------|---------|
| TMC 5m (Raw Observed)  | N/A       | N/A  | 45%        | 18.0%        | 4018.7    | 8ms     |
| Bicubic 1m             | 23.1 dB   | 0.74 | 58%        | 11.8%        | 3066.9    | 12ms    |
| EDSR 1m                | 26.4 dB   | 0.83 | 75%        | 6.2%         | 2517.0    | 185ms   |
| SwinIR 1m              | 27.1 dB   | 0.85 | 79%        | 5.3%         | 2368.9    | 420ms   |
| LunarSR 1m (LUNARSR)   | 24.5 dB   | 0.81 | 86%        | 3.1%         | 2115.1    | 10909ms |

**SCIENTIFIC LIMITATION NOTICE:** Super-resolved imagery is an estimated higher-resolution representation derived from lower-resolution source imagery. It must not be interpreted as direct physical observation. Slope, physical terrain roughness, local relief and elevation-dependent hazards require valid elevation data (DEM) or a separately validated terrain reconstruction method.